Project Name: Human Emotion Detection from Voice

Introduction

This is a Web browser-based application to detect the emotion from human voice in English speech. It can be useful for clinical diagnosis of anxiety, depression, or stress, helping professionals monitor their patients. It can be advantage in a field of forensic science as well.

URL: https://huggingface.co/spaces/aishwaryaIngole/voice-emotion-detection

Abstract

Human Emotion Detection from voice is a critical component in developing emotionally intelligent systems for applications ranging from mental health monitoring to criminal forensics. This project presents a robust pipeline for the classification of eight distinct emotional states—neutral, calm, happy, sad, angry, fearful, disgust, and surprised—using the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS).

The methodology focuses on high-fidelity feature engineering, extracting 180+ features including **Mel-Frequency Cepstral Coefficients (MFCCs)**, **Chroma STFT**, and **Mel-Spectrograms**. To capture the dynamic prosody of human speech, first and second-order **polynomial deltas** (velocity and acceleration) were integrated. To address real-world environmental variability, a specialized preprocessing layer was implemented featuring **fixed-rate resampling (22,050 Hz)**, **vocal pre-emphasis filtering, supressing background noise**, and **automated silence trimming**.

The classification was performed using a **Stratified Random Forest** model with 500 estimators, achieving a validated **accuracy of 93.75%**. The system was deployed via a **Gradio-based web interface**, enabling real-time vocal recording, emotional trend visualization, and session data logging.

Tools Used

This project utilized a specialized stack of audio processing, machine learning, and interface tools.

1. **Programming Language Python** the core language used for scripting, data processing, and model training due to its extensive support for AI libraries.
2. **Audio Processing & Feature Extraction**
    a. **Librosa:** The primary library for music and audio analysis. Used for extracting **MFCCs**, **Chroma**, **Mel-Spectrograms**, and performing **Vocal Pre-emphasis** and **Silence Trimming**.
    b. **Soundfile:** Used as the high-speed backend for reading .wav files and handling metadata like native sample rates.
    c. **Resampy:** Utilized by Librosa to perform high-quality **Kaiser-window resampling** (22,050 Hz) to ensure consistency across different microphones.
    d. **NoiseReduce:** The library is used to suppress the background noise in real time recordings.
3. **Machine Learning & Data Science**
    a. **Scikit-Learn (Sklearn):** The backbone of the intelligence layer.
        i. Used **RandomForestClassifier** (500 estimators) for the main engine.

        ii. Used **StandardScaler** for feature normalization.
        iii. Used **LabelEncoder** to map emotional strings to numerical categories.
    b. **Pandas:** Used for structured data management and generating the session_log.csv for user downloads.
    c. **NumPy:** Used for high-performance matrix operations and "stacking" the 180+ feature vectors.
4. **Visualization & Deployment**
    a. **Gradio:** Used to build the **Graphical User Interface (GUI)**. It handles the browser-based microphone recording and the live prediction display.
    b. **Matplotlib:** Used to generate the **Emotional Trend Graph** which visualizes how a user's mood shifts over a session.
    c. **Joblib:** Used for **Model Persistence**—saving the trained 93.75% model to a .pkl file for instant loading without retraining.
    d. **Hugging Face Spaces:** The cloud platform used for **Global Deployment**, hosting the model via Git LFS (Large File Storage).
5. **Version Control Git & Git LFS:** Used to manage code versions and handle the large binary model files (.pkl) during the deployment process.

Operational Stages

1. **Data Acquisition** The system was trained using the **Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS)**. This dataset contains 7,356 files from 24 professional actors. For this project, the **speech-only** subset was utilized, encompassing eight emotional labels: *neutral, calm, happy, sad, angry, fearful, disgust, and surprised*. The dataset was selected for its high internal validity and gender-balanced recordings.

2. **Signal Preprocessing** To ensure the model generalizes to real-world microphone inputs and remains independent of recording volume, a rigorous preprocessing layer was implemented using the Librosa and Soundfile libraries:
    a. **Resampling:** All audio signals were standardized to a fixed sample rate of **22,050 Hz** to maintain consistent frequency bins across the training and testing sets.
    b. **Vocal Pre-emphasis:** A high-pass filter was applied to the raw signal to amplify high-frequency components, sharpening the vocal cords' spectral clarity.
    c. **Silence Trimming:** A decibel-based threshold (25dB) was used to remove leading and trailing silences, ensuring the model analysed only active speech.
    d. **Amplitude Normalization:** Peak-normalization was applied to the waveforms to prevent "loudness" from being misclassified as "anger."
    e. **Nosie Suppression:** The background noise while real-time recording is removed.
3. **Feature Engineering** The core of the system's high accuracy lies in its multi-dimensional feature vector. Over **180 acoustic features** were extracted per sample:
    a. **Timbral Features:** 40 Mel-Frequency Cepstral Coefficients (MFCCs) were extracted to capture the shape of the vocal tract.
    b. **Temporal Dynamics (Deltas):** First-order (velocity) and second-order (acceleration) MFCC derivatives were calculated to capture the **prosody** and rate of change in emotional expression.
    c. **Pitch & Harmonic Features:** 12-element Chroma STFT vectors were used to identify tonal shifts, while Mel-Spectrograms captured energy distribution across frequency bands.

4. **Model Architecture and Training**
    a. **Data Scaling:** All features were transformed using a **StandardScaler** to ensure a mean of zero and a standard deviation of one, preventing high-magnitude features (like Mel energies) from dominating the model.
    b. **Classification:** A **Random Forest Classifier** was selected due to its robustness against overfitting in high-dimensional feature spaces. The model was configured with **500 decision trees** to provide stable "ensemble" voting.
    c. **Validation Strategy:** An **80/20 Stratified Train-Test Split** was employed. Stratification ensured that the distribution of emotions (e.g., the 96 'Surprised' samples vs. 192 'Happy' samples) remained identical in both sets, leading to a verified **accuracy of 94.10%**.
5. **Deployment Interface** The finalized model, scaler, and label encoder were serialized using Joblib. A real-time interface was developed using Gradio, enabling users to record audio through a web browser, which is then processed through the identical preprocessing and feature extraction pipeline for instant prediction.

Conclusion

Through a vast dataset, we successfully achieved 93.75% accuracy. But in real time predictions, we have observed a dip in this due to multiple reasons. Dataset used is in English with just 2 sentences - "Kids are talking by the door" and "Dogs are sitting by the door" and recorded in silent studio. The actors have a specific Canadian accent. Real time recordings are affected by microphone quality, background noise, normal voice unlike acted in dataset, and with different English words and hence the result despite taking care of all such parameters with scaling, frequency matching, normalization etc. Sometimes the predictions can be wrong. Necessity for multi-lingual dataset was felt while testing and therefore it offers a possibility of many corrections to be applied in future.